Introduction to within-host modeling of viral dynamics & viral immunology

1. Introduction

Physics is the scientific discipline were the states and dynamics of natural phenomena are described and modeled using mathematics. You all know the famous equations formulated by Isaac Newton and colleagues.

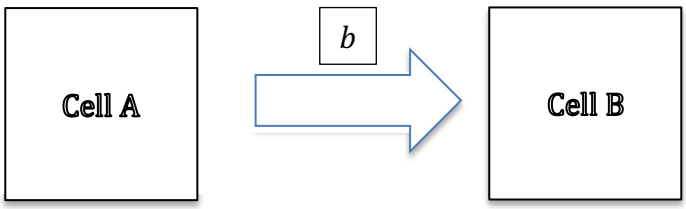
In this syllabus, I will briefly detail how mathematical modeling can be applied to obtain biological insights which were otherwise not obtainable. We will start with a general overview of modeling techniques and this will be followed by different within-host modeling examples.

2. Types of modeling approaches

In general, we can distinguish two major types of modeling techniques: ordinary differential equation (ODE)-based modeling (ODEM) & agent-based modeling (ABM).

2.1. ODE-based modeling

In this approach we typically define different compartments to allocate the different biological entities into. Let's say we are modeling cell dynamics, we can note the dynamics between **compartments** like



Here, we state that cells go from Cell type A ($Cell_A$) to Cell type B ($Cell_B$) at a rate *b*. In mathematical terms, we can write

$$\begin{cases} \lim_{\Delta t \to 0} \frac{\Delta Cell_A}{\Delta t} = \frac{dCell_A}{dt} = -b * Cell_A \to Cell_A = C_{st} * e^{-b*t} \\ \frac{dCell_B}{dt} = +b * Cell_A = b * C_{st} * e^{-b*t} \end{cases}$$

These equations describe the dynamics between $Cell_A$ and $Cell_B$ and we can use R/Matlab/Python or speciliazed programs like Monolix to fully solve these equations.

2.2. AB-modeling

You should note that in ODEM we typically don't account for the duration of a cell in a specific compartment. Once the cell has entered the compartment it becomes one with all other cells present in the compartment. Moreover, the dynamics between cell compartments is defined by exponential distributions (which is quite typical for biological dynamic systems).

If we want to specify the cellular dynamics at a more cell-individual level we need to apply another kind of modeling, i.e. agent-based modeling (=individual-based modeling). In ABM, every cell will receive an unique index and a unique set of characteristics. We will track the cell throughout its life in the individual and at every time step we will update the cell's characteristics (e.g. going from naïve T-cell to effector memory T-cell or memory T-cell, see Figure 1). This approach will allow a full and realistic view of the immune system. However, clearly, this approach will

be much more computationally intensive than the ODEM approach and must be undertaken on a supercomputer.

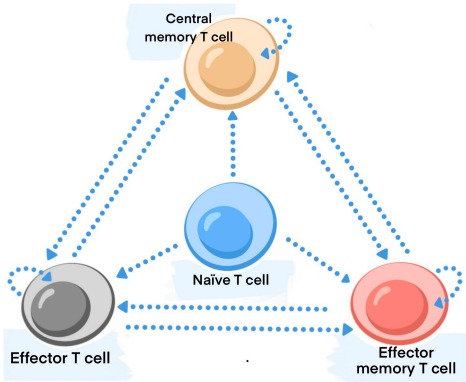


Figure 1: Basic T cell differentiation model

2.3. Deterministic vs stochastic modeling

Finally, within the previous frameworks one can choose between more deterministic approaches or more stochastic approaches with the difference obviously concerning the consideration of random processes in the models.

3. Elementary Viral Dynamics

Imagine a population of uninfected cells U, a population of infected cells I and a population of circulating viruses v (see Figure 1).

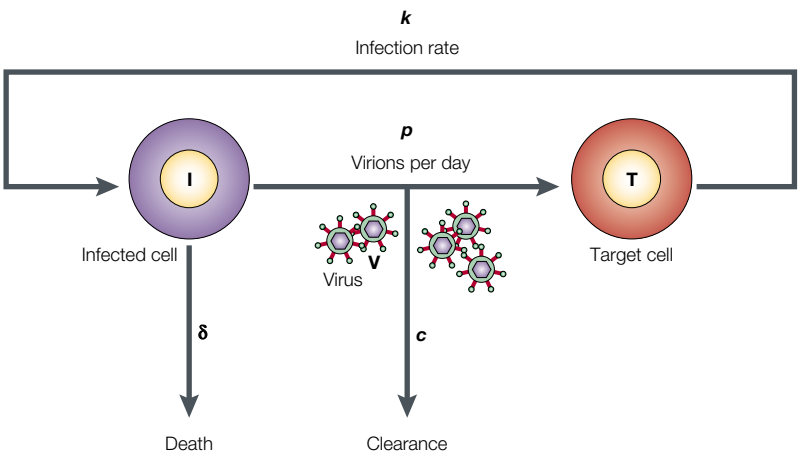


Figure 2: Basic model of viral infection (Perelson, Nature Reviews Immunology, 2012)

We can construct the following set of ordinary differential equations:

$$\begin{cases} \dfrac{dU}{dt} = B - \mu_U \cdot U - \beta \cdot U \cdot v \\ \dfrac{dI}{dt} = \beta \cdot U \cdot v - \mu_I \cdot I \\ \dfrac{dv}{dt} = k \cdot I - \mu_v \cdot v \end{cases}$$

with B = the rate of new uninfected cells, k = the rate of virus production in infected cells and $\mu_{U,I,v}$ = the respective removal rates.

As an example of the strength of mathematical modeling, we will briefly discuss the use of antivirals against HIV. Reverse transcriptase (RT) inhibitors block the ability of HIV to infect a cell, whereas protease inhibitors (PI) cause the production of non-infectious viral particles (v_{NI} in contrast to v_I, with $v = v_I + v_{NI}$). We can adapt our set of equations to

$$
\begin{cases}
\dfrac{dU}{dt} = B - \mu_U \cdot U - (1-\varepsilon_{RT}) \cdot \beta \cdot U \cdot v_I \\
\dfrac{dI}{dt} = (1-\varepsilon_{RT}) \cdot \beta \cdot U \cdot v_I - \mu_I \cdot I \\
\dfrac{dv_I}{dt} = (1-\varepsilon_{PI}) \cdot k \cdot I - \mu_v \cdot v_I \\
\dfrac{dv_{NI}}{dt} = \varepsilon_{PI} \cdot k \cdot I - \mu_v \cdot v_{NI}
\end{cases}
$$

4. Elementary Immunology

Ever since the beginning of mankind, and even before in older phylogenetic branches, infections have caused significant morbidity and mortality, and disrupted societies. Fortunately, encounters with pathogens such as viruses, bacteria and parasites do not progress without a fight, and we can count on a well-equipped system in our body, called the **immune system**. I will give here a brief and sometimes simplistic view on how the immune system works for a better understanding of the project proposal.

The immune system is a multi-layered organ that combats pathogens at different locations and on different time points. The first layer is the **innate immune system**, which consists of physical barriers such as epithelia, but also secreted molecules that aim to destroy or slow-down the infectious pathogens. Innate immunity also consists of cells that can recognize pathogens upon encounter in a rather unspecific matter, although receptors and other mechanisms exist, which may recognize more specific parts of pathogens. Macrophages, natural killer cells and neutrophils construct the typical first line cellular barrier against infections. These cells can protectively interact with infected host cells as well as circulating pathogens, for example by excreting toxic molecules.

The **adaptive immune system** represents the specific response against pathogens. The actors of the adaptive immune system can recognize specific parts of the pathogen-derived proteins. These specific parts are called epitopes or antigens and consist mainly of 9-20 amino acid long peptides. Once a pathogen breaches our physical barriers so called

antigen-presenting cells (APC) will pick up the pathogens and process it into epitopes. The APC bind the epitopes to major histocompatibility complex (MHC) molecules and subsequently relocate the epitope-MHC on the cell surface, ready for recognition. Next, the APC will traffic to lymph nodes where they will encounter the actors of the adaptive immune system: B-cells and T-cells.

The B-cells and T-cells are both part of adaptive immunity but they represent two distinct branches called humoral immunity and cellular (mediated) immunity (CMI), respectively. T-cells carry T-cell receptors (TCR) on their cell surface that are capable of recognizing specific epitope-MHC presented by an APC on its cell surface. Naïve B-cells can become activated when the B-cell receptor (BCR) recognizes an epitope on a pathogen. Interestingly, each human possesses millions of distinct BCR and TCR that enable us to recognize an extreme broad repertoire in pathogenic variation.

After BCR-epitope binding, the **B-cell** takes up the pathogen via receptor-mediated endocytosis and degrades it in the lysosome after which the epitopes will be presented on the MHCII of the B-cell that can then interact with the TCR of a T-cell. After the T-cell help, the B-cells become activated and will clonally expand in the germinal center of the lymph nodes. Finally, these B-cells will become antibody-secreting cells. The antibodies, in our context first IgM and next IgG, will circulate in the blood stream where they can recognize the circulating pathogens and neutralize them. Activated B-cells are capable of differentiating into long-living plasma cells that traffic towards the bone marrow, where they will reside for several decades. These plasma cells continue to produce

antibodies without the need of an antigenic stimulus. B-cells can also differentiate into memory B-cells that remain in lymphoid organs after the first infection has resolved. These memory B-cells allow for a much more rapid response in antibody production when a second encounter with the pathogen occurs.

The kinetics of antibodies following first infection is well known: after first encounter (eliciting a **primary immune response**) the antibodies (from now on we will primarily focus on IgG) show an increase in titer after several weeks. Next, the antibody titer will have a rapid decay phase, followed by a rather steady level for several years.
When a second encounter with the pathogen occurs, a **secondary immune response** is elicited, which will cause an even more rapid increase in IgG followed again by a rapid decay phase and steady level.

5. Elementary CMI

Upon viral infection, adaptive immunity will not simply watch and wait, but will initiate a response directed at the virus in question. Once again, we can formulate a set of differential equations describing an elementary cellular immune response:

$$\frac{dv}{dt} = r \cdot v - c \cdot v \cdot E,$$
$$\frac{dN}{dt} = \sigma + r_N N - a_N N - \mu_N N,$$
$$\frac{dE}{dt} = a_N N + a_M M + pE - mE - \mu_A E$$
$$\frac{dM}{dt} = mE + r_M M - a_M M - \mu_M M$$

The first equation describes the evolution of a virus (v) after introduction to the system (for e.g. the human body). The virus replicates at a rate r (note that we have defined a new replication rate), but is simultaneously reduced by the protective actions by effector cells (E) with an efficacy rate c (for clearance). The second equation shows that naïve cells (N) are continuously formed at a rate σ, renewed at a rate r_N, but reduced by the activation towards effector cells at a rate a_N and death at a rate μ_N. The third equation denotes that effector cells are generated via naïve cells, but also by the transformation of memory cells (M) at a rate a_M and by the proliferation of effector cells at a rate p. Effector cells are reduced by the transformation towards memory cells at a rate m, and by death at a rate μ_A. In the final equation we see that memory cells are generated via effector cells, renewal at a rate r_M, but are reduced by the transformation towards effector cells at a rate a_M, and by death at a rate μ_M.

We note that we can adapt in the N equation the activation term $-a_N*N$ to $-a_N*v*N$ to account for a naïve to effector activation rate that depends on the viral count. However, in order to account for saturation effects regarding the viral count, we can modify the activation term to $-a_N \cdot \frac{v}{h+v} \cdot N$ or alternatively $-a_N \cdot \frac{1}{h+e^{-v}} \cdot N$, where h defines the saturation function.

6. Elementary HMI

Not only cellular immunity is of importance in the defence against pathogens, but also humoral immunity. Broadly spoken, we can design a mathematical model in which antibodies are produced after vaccination or infection by B-cells:

$$\frac{dBcell}{dt} = a \cdot T(T = 1\ if\ t < h\ and\ 0\ if\ not) - \mu_B \cdot Bcell$$

$$\frac{dAB}{dt} = p \cdot Bcell - \mu_B \cdot AB$$

with *Bcell* the number of B-cells at a certain time point, *a* the constant generation of B-cells during an activation period *h*, AB the number of antibodies, and *p* the AB production rate by B-cells.

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Please note that every year we have bachelor and master-thesises for both biomedical students (thus no direct equation modeling, mainly experiments in the lab) and physics students (ODE/AB modeling).

My lab is unique in the sense that we generate our own data and perform the subsequent data modeling to infer new insights related to within-host dynamics.